

HARDWARE CAPSTONE

Topic: Design a Useful Gadget (Grade 4 Science & Technology)



What is a Gadget?

A gadget is a small tool or device that has a specific, useful purpose. Gadgets help us perform daily activities more easily, quickly, and efficiently!

Pressure Cooker

Cooks food much faster using steam pressure and fuel! It saves time and energy in the kitchen.

Pencil Box

Keeps pencils, erasers, rulers, and sharpeners perfectly organized in neat compartments so they don't get lost!

Safety Pin

Holds fabric together securely and safely—no more falling badges or loose clothes!



Gadgets Solve Real-World Problems!

Every single gadget you see around you was invented because someone noticed a problem and wanted to find a better way to solve it.

Idli Stand

Stacks steaming plates upwards to cook many idlis all at once without taking up extra space on the stove!

Cycle Pump

Fills bicycle tires easily with air so your ride is always smooth wherever you go!

Dabba Delivery System

Keeps delicious homemade food perfectly warm while it is being delivered across large, busy cities!



The Design Loop

Designers follow a special repeating cycle to create amazing new tools. To turn your ideas into a real gadget, follow these three crucial steps:

THINK



MAKE



SHARE

Keep looping, testing, and improving your design to make your gadget even better!

Meet an Amazing Indian Innovator

Mansukhbhai Prajapati's Mitticool Clay Fridge

Mansukhbhai noticed a big problem: many families living in rural villages did not have reliable electricity, so their fresh food, milk, and vegetables would spoil quickly. Using his traditional pottery skills, he invented the **Mitticool Fridge** out of natural clay! It keeps food fresh and cool completely without electricity. What an incredible solution!

Problem Solving: Ravi's Leaky Bottle

Let's look at a problem close to home. Ravi's water bottle leaks and wastes precious water every single day at school. The water drips inside his bag and ruins his schoolbooks!

- **What is the problem?** The bottle leaks and ruins books.
- **How can he fix it?** What simple materials (like rubber bands, tape, or plastic caps) could he use to design a leak-stopping gadget?

Your Turn: My Problem Statement

Think about a problem you see around your home or school. It could be a squeaky door, a heavy school bag that hurts when climbing stairs, or a tap that continuously drips.

1. I noticed a problem when:

.....

2. The problem is:

.....

3. This bothers/affects:

.....

4. I think a new gadget could help because:

.....

Next Step: Get Ready to Build!

Soon, you will build a prototype of your gadget. Start collecting safe, everyday materials like clean cardboard boxes, tape, strings, bottle caps, and recycled items to bring your brilliant idea to life!



STUDENT PRACTICE & REFLECTION WORKSHEET

Grade 4 Capstone Challenge: Review and Reinforce Your Design Skills



Practice Questions (No Answers - Try Your Best!)

Answer the following questions based on what you have learned in this lesson about gadgets and the design loop.

1. What is the main definition of a **gadget**? Use your own words to describe it.

2. Name three household gadgets mentioned in this lesson and explain what they are used for.

3. Explain how an **Idli Stand** solves a space-saving problem in the kitchen.

4. What are the three core steps of the **Design Loop**? List them in the correct order.

5. Why is it important to repeat or keep looping through the steps of the Design Loop instead of stopping after your first try?

6. In the case study of innovator Mansukhbhai Prajapati, what real-world problem inspired him to create the **Mitticool Fridge**?

7. What unique material did Mansukhbhai use to build his fridge, and what makes it environment-friendly?

8. Describe the specific problem Ravi faces with his water bottle. Who or what does this problem affect?

9. If you were helping Ravi, what are two everyday materials you would recommend he use to create a simple prototype to fix the leak?

10. Look around your classroom or home right now. Identify one thing that does not work well or could be improved. Write it down as a problem.

11. What is a **prototype**? Why do designers make a sample before building the final product?

12. Why is the **"SHARE"** step in the design loop vital when creating a gadget for other people?

 3-2-1 Exit Ticket Challenge:

- 3)** Name 3 gadgets you learned about today.
- 2)** List 2 steps of the design loop you remember.
- 1)** Write 1 feeling word that describes how you feel about becoming a designer today!